

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 16: Loci (Locus and its Constructions)

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Loci in a triangle

1. Statement 1: The point equidistant from three non-collinear points D , E and F is the circumcentre of $\triangle DEF$. Statement 2: The incentre of a triangle is the point where the bisectors of its angles meet. (a) both true (b) both false (c) 1 true, 2 false (d) 1 false, 2 true **ICSE 2025, 1 mark**

SOLUTION

A point equidistant from the three vertices lies on the perpendicular bisectors of all three sides. That point is the circumcentre, so statement 1 is true.

The angle bisectors meet at the point equidistant from the three sides, which is the incentre, so statement 2 is true.

Answer: (a)

2. The circumcentre of a triangle is the point that is: (a) equidistant from the three sides (b) equidistant from the three vertices (c) the meeting point of the three medians (d) the meeting point of the three altitudes **ICSE 2024, 1 mark**

SOLUTION

The circumcentre is the centre of the circle through the three vertices, so it is the same distance from each vertex.

Answer: (b)

2 Construction questions

3. Using ruler and compasses, construct $\angle ABC = 90^\circ$ with $AB = 6$ cm and $BC = 8$ cm. Construct
 (a) the locus of points equidistant from B and C (b) the locus of points equidistant from A and B
 (c) the locus of points at the fixed distance OA from O , where O is the point common to (a) and
 (b) (d) the locus of points equidistant from BA and BC . **ICSE 2024, 4 marks**

SOLUTION

Draw $BC = 8$ cm, construct a right angle at B and cut off $BA = 6$ cm.

- (a) Draw the perpendicular bisector of BC .
 (b) Draw the perpendicular bisector of AB .
 (c) The two bisectors meet at O . With centre O and radius OA , draw a circle. It passes through A , B and C .
 (d) Draw the bisector of $\angle ABC$.

Check: O is the mid-point of the hypotenuse AC , and $AC = \sqrt{6^2 + 8^2} = 10$ cm, so the radius OA should measure 5 cm.

Answer: the loci are two perpendicular bisectors, a circle of radius 5 cm and the bisector of $\angle ABC$

4. Using ruler and compasses, draw a circle of radius 4.5 cm and a chord $AB = 6$ cm. (a) Construct the locus of points equidistant from A and B , and mark the point D where it meets the circle. (b) Join AD , construct the locus of points equidistant from AD and AB , and mark the point C where it meets the circle. (c) Join BC and CD and measure CD . **ICSE 2020, 4 marks**

SOLUTION

- (a) The locus is the perpendicular bisector of AB . It passes through the centre and meets the circle at two points; take D as the point on the far side of the centre from AB .
 (b) The locus is the bisector of $\angle DAB$. It meets the circle again at C , which is the mid-point of the arc BD .
 (c) Join BC and CD . Because AC bisects $\angle DAB$, the chords BC and CD are equal.

Answer: CD measures about 5.1 cm

5. Using ruler and compasses, construct a semicircle on the diameter $BC = 7$ cm. Locate the point A on the semicircle that is equidistant from B and C . Complete the cyclic quadrilateral $ABCD$ with D equidistant from AB and BC . Measure $\angle ADC$. **ICSE 2019, 4 marks**

SOLUTION

Draw $BC = 7$ cm and its perpendicular bisector, meeting BC at O . With centre O and radius 3.5 cm draw the semicircle.

The perpendicular bisector is the locus of points equidistant from B and C . It meets the semicircle at A .

The bisector of $\angle ABC$ is the locus of points equidistant from AB and BC . It meets the semicircle at D . Join AD and DC .

Check: $\angle BAC = 90^\circ$ and $AB = AC$, so $\angle ABC = 45^\circ$. Opposite angles of a cyclic quadrilateral add up to 180° .

Answer: $\angle ADC = 135^\circ$

These questions are based on questions asked in past board examinations. They have been reworded by MathSetu, the figures are redrawn, and the solutions are MathSetu's own. MathSetu is an independent study resource and is not affiliated with or endorsed by CBSE, CISCE, NCERT or any publisher.